

➤ General Instructions:

- The submission due date of this assignment is **Monday(28th March 2026) 11:59 PM**
- Write a report (i.e. in a word file) that illustrates your main solution steps including the best fitness values and the average values, plotted over generations (with and without elitism)
- Zip your code and the report in a file entitled **[YourName_YourID_AssignmentNumber]**
- This assignment should be delivered and discussed INDIVIDUALLY

➤ Requirements:

In light of Assignment #2, solve the following optimization problem:

$$\text{Maximize } F(x_1, x_2) = 8 - (x_1 + 0.0317)^2 + (x_2)^2, \text{ where } -2 \leq x_1, x_2 \leq 2$$

1. Initialize the population using real encoding, where $R_{\max}$ and $R_{\min}$ are two real numbers
 $\text{pop} = \text{init_pop}(\text{pop_size}, R_{\max}, R_{\min})$
2. Apply arithmetic crossover operator as follows
 $\text{two_children} = \text{arithmetic_cross}(\text{two_parents}, P_{\text{cross}} = 0.6)$
3. Build a function to apply Gaussian mutation with a fixed standard deviation, your function may look like...
 $\text{new_individual} = \text{gaussian_mutate}(\text{individual}, \text{sigma} = 0.5, p_{\text{Mut}} = 0.05, R_{\max}, R_{\min})$
4. In the previous two assignments, the selection process was done using Roulette wheel selection, in this assignment you are asked to apply (**Tournament Selection**) with two different k values (very small and very large). Compare the results of both.
 $\text{Selected_Individuals} = \text{tournament}(\text{pop}, k)$
5. All other implementation settings in the previous assignment should remain the same (i.e. population size, number of generations, elitism with elite size = 2)
6. Do you have any improvements over Assignment#2's results? Which of the above operators is the most effective one (Selection, Crossover, or Mutation) in improving the results?

BEST OF LUCK!